

Honest inference for rare probabilities of causation

CausalSmith Stage -1 proposal

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Locked parameters. The row index is n , the covariate dimension p is fixed, and $O_i = (X_i, A_i, Y_i)$ is row-wise i.i.d. from P_n . The target is $\theta(\tilde{P}_n) = \mathbb{P}\{Y(0) = 0 \mid A = 1, Y(1) = 1\}$ under consistency, ignorability, monotonicity, and overlap. The observed-law functional is

$$\Theta_{\text{PN}}(P_n) = 1 - \frac{\mathbb{E}_{P_n}[e_n(X)m_{0,n}(X)]}{\mathbb{E}_{P_n}[e_n(X)m_{1,n}(X)]}, \quad d_n = \mathbb{E}_{P_n}[AY].$$

The regression $m_{0,n}$ is fitted only on control training observations by clipped tensor-product B-spline ERM.

Abstract

The proposal studies inference for PN and PS when their positive factual conditioning events can be rare. It retains all four treatment–outcome cells and the random-covariate component of the marginal causal null in a randomized empirical-support profile likelihood ratio. The finite profiling experiment has randomized coverage at least $1 - \alpha$; a zero-denominator closure is used only for least-favorable calibration, and a marked-Poisson limit is defined under explicit marked-intensity convergence. A primitive Hölder calculation isolates the continuous-law Gaussian regime in which spline learning error is negligible at the rare-event scale. A monotone causal submodel supplies a matching effective-count lower bound. Whether a higher-order likelihood transfer bridges the remaining diverging-count region is an explicit open question.

1 Introduction

Probabilities of necessity and sufficiency give a causal attribution interpretation to an observed harmful or protective event. Their conditioning events can be rare in stroke attribution and climate-extreme attribution.

Tian and Wu (2025) derive monotonicity-based influence functions and regular semiparametric inference for PN and PS. Their regular-law benchmark does not address triangular rare-event sequences, randomized finite-count inversion, or marked-Poisson limits. The question is how honest confidence-set diameter changes with nd_n when nuisance functions are learned.

The immediate consumer is rare-event attribution: a fixed-support analysis can use randomized discrete calibration when the factual count is small, whereas a sufficiently smooth continuous analysis can use the Gaussian endpoint. This distinguishes lack of information from failure of a Wald approximation.

- Theorem 1: finite empirical-support profiling has coverage at least $1 - \alpha$ and a marked-Poisson boundary. Gap: Huwang (1995) and McGrath–Burke (2024) concern binomial intervals, not learned causal score cells. Consumer: fixed-support attribution.
- Theorem 2: the Hölder/spline condition identifies an attainable Gaussian PN/PS regime. Gap: Tian–Wu (2025) is fixed-law. Consumer: rare stroke attribution.
- Theorem 3: every uniformly honest set has diameter at least order $\min\{1, (nd_n)^{-1/2}\}$. Gap: Gleser–Hwang (1987) does not certify a monotone PN submodel. Consumer: an information bound for attribution designs.
- Open-ended Question 1: can higher-order empirical likelihood attain the full frontier outside the first-order Gaussian regime? Consumer: a single procedure for rare climate and clinical attribution.

2 Related work

Tian and Wu (2025) is the regular PN/PS comparator. Cai and Kuroki (2006) study regular variance estimators for probabilities of causation. Chernozhukov et al. (2018) supplies the cross-fitting architecture, and Chen (2007) supplies sieve-rate tools. Mikusheva and Sun (2022) develop identification-robust test inversion for many weak instruments, including jackknife Anderson–Rubin methodology; its linear-IV experiment is distinct from rare PN/PS, learned nuisances, empirical-support profiling, and marked-Poisson limits. Huwang (1995) documents failure of a normal binomial interval, while McGrath and Burke (2024) compares rare-binomial coverage and relative margins. Gleser and Hwang (1987) motivates the diameter lower-bound tem-

plate. Hannart and Naveau (2018) and Kiriliouk and Naveau (2020) motivate the climate-attribution consumer.

3 Formal setup

Let $e_n(x) = \mathbb{P}(A = 1 \mid X = x)$ and $m_{a,n}(x) = \mathbb{E}(Y \mid A = a, X = x)$. Under the causal assumptions,

$$\theta(\tilde{P}_n) = \Theta_{\text{PN}}(P_n), \quad \vartheta(\tilde{P}_n) = 1 - \frac{\mathbb{E}[(1 - e_n(X))(1 - m_{1,n}(X))]}{\mathbb{E}[(1 - e_n(X))(1 - m_{0,n}(X))]}.$$

For $t \in [0, 1]$, the PN score is

$$\psi_{t,n}(O; \eta) = A\{Y - m_0(X)\} - \frac{e(X)}{1 - e(X)}(1 - A)\{Y - m_0(X)\} - tAY.$$

4 Assumptions

The full-data extension obeys consistency, $(Y(0), Y(1)) \perp A \mid X$, and $Y(1) \geq Y(0)$. The observed law has $\varepsilon \leq e_n(X) \leq 1 - \varepsilon$, $c_X \leq f_{X,n}(X) \leq C_X$, $e_n \in \mathcal{H}^{\beta_e}(L_e)$, $m_{0,n} \in \mathcal{H}^{\beta_0}(L_0)$, $0 < d_n \leq \bar{d}$, and $d_n^{\text{PS}} = \mathbb{E}[(1 - A)(1 - Y)] > 0$. Here $p, \beta_e, \beta_0, L_e, L_0$ are fixed. The Gaussian sub-class additionally requires true-target score nondegeneracy and

$$d_n \gg \max \left\{ (\log n/n)^{2s_0}, n(\log n/n)^{2(s_e + s_0)} \right\}, \quad s_j = \frac{\beta_j}{2\beta_j + p}.$$

5 Main results

Theorem 1 (Finite empirical-support inversion). *At labelled observed covariates, profile a four-cell law with empirical weights $w \in \Delta_n$, cell probabilities $q_{i,ay}$, and the marginal null $\sum_i w_i e_i[(1 - t)u_i - m_{0,i}] = 0$. PN is parameterized only by laws with $d(\xi) > 0$; the compact $d(\xi) = 0$ closure is retained solely for least-favorable calibration. Randomized likelihood-ratio inversion has coverage at least $1 - \alpha$ under every positive-denominator null law in this finite experiment. If $N_{11} = 0$, the reported set is $[0, 1]$. Along a sequence with $nd(\xi_n) \rightarrow \lambda$ and convergence of the joint rare-cell mark intensity and common-cell profile process, the likelihood-ratio experiment converges to the defined marked-Poisson experiment. The treatment/outcome swap gives PS when $d_n^{\text{PS}} > 0$.*

Theorem 2 (Attainable Gaussian recovery). *Use fixed-fold clipped tensor-product B-spline ERM with $K_{j,n} \asymp (n/\log n)^{1/(2\beta_j+p)}$ knots. Without invoking monotonicity, the observed-law learning calculation gives*

$$O_{P_n} \left(\frac{a_{0,n}}{\sqrt{n}} + a_{e,n} \sqrt{\frac{\mathbb{E}[m_{0,n}(X)]}{n}} + a_{e,n} a_{0,n} \right), \quad a_{j,n} = (\log n/n)^{s_j}.$$

The causal specialization uses consistency, ignorability, monotonicity, and overlap to obtain $m_{0,n}(x) \leq m_{1,n}(x)$ and $\mathbb{E}[m_{0,n}(X)] \leq d_n/\varepsilon$. On the displayed Gaussian sub-class, the likelihood transfers at scale $\sqrt{nd_n}$ and the expected diameter is $O((nd_n)^{-1/2})$ under $P_n^{\otimes n} \otimes \text{Unif}$. The fixed- d_n special case recovers the Tian–Wu regular semiparametric benchmark; the swapped claim requires $d_n^{\text{PS}} > 0$.

Theorem 3 (Causal effective-count lower bound). *For every measurable uniformly honest confidence set D_n ,*

$$\liminf_n \sup_{P_n \in \mathcal{M}_n} \frac{\mathbb{E}_{P_n^{\otimes n}}[\text{diam}(D_n)]}{\min\{1, (nd_n)^{-1/2}\}} > 0.$$

The least-favorable submodel has $X \sim \text{Unif}([0, 1]^p)$, $A \sim \text{Bernoulli}(1/2)$, and monotone types $(1, 1), (0, 1), (0, 0)$ with probabilities $2(1-t)d_n, 2td_n, 1-2d_n$. Hence its observed four cells are $(d_n, (1-t)d_n, 1/2-d_n, 1/2-(1-t)d_n)$, which verifies the structural causal and Hölder membership conditions. Thus $nd_n \rightarrow \lambda < \infty$ forces nonshrinking expected diameter.

Open-ended Question 1 (All-regime transfer). *Can a higher-order orthogonal empirical-likelihood expansion jointly control sieve approximation, random-covariate empirical weights, and score marks at the rare-event scale, so that the same profile inversion has honest coverage and diameter $O\{\min(1, (nd_n)^{-1/2})\}$ over every diverging-count sequence outside the displayed Gaussian sub-class?*

6 Sanity-check corollaries

With no covariates in the lower submodel, condition on $M = N_{11} + N_{01}$. Then $N_{01} \mid M \sim \text{Binomial}(M, q_t)$ with $q_t = (1-t)/(2-t)$. For counts $(2, 1, 8, 9)$, $M = 3$, $\hat{q} = 1/3$, and $\hat{t} = 1/2$. Enumerating $j = 0, 1, 2, 3$ gives the randomized upper-tail likelihood-ratio acceptance rule with no zero likelihood denominator for $0 < d_n < 1/2$ and $0 < t < 1$.

7 Proof-sketch route

The finite theorem follows from finite-support least-favorable randomization and Poisson convergence. The Gaussian theorem chains spline approximation, cross-fit score algebra, monotone rare-event localization in expectation, and a likelihood transfer/LAN expansion. The lower theorem chains certified structural membership, a local pair separated by $h \min\{1, (nd_n)^{-1/2}\}$, product Kullback–Leibler control, and testing-to-diameter reduction. The open question requires a new higher-order replacement for the first-order rare-scale remainder.

8 Honest scope flags

Finite-sample coverage is only for the finite empirical-support profiling experiment. Its PN parameter is defined only for positive-denominator laws; the $d(\xi) = 0$ closure is a least-favorable calibration device, and the marked-Poisson statement requires the stated convergence of marks and common-cell likelihoods. PS claims require $d_n^{\text{PS}} > 0$. Over continuous Hölder laws, shrinking diameter is asserted only in the stated Gaussian regime. First-order fixed-smoothness learning does not cover every sequence with $nd_n \rightarrow \infty$; the higher-order bridge is open.

9 Iteration trail

The carried-forward revision retains the four-cell empirical-support likelihood and removes conditional-on-covariate calibration. It separates finite-count validity from the continuous-law Gaussian transfer, derives the spline remainder from primitive conditions, removes pointwise rare-event localization, and corrects the causal lower-submodel membership and bibliography records.

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